

Desmond Coles

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SUMMARY

Mathematician and recent PhD graduate from the University of Texas at Austin, with background in abstract algebra and combinatorics. Excited to use mathematical rigor and problem solving to build reliable, maintainable, and useful products.

EDUCATION

The University of Texas at Austin, PhD in Mathematics	3.94 GPA	2025
The Ohio State University, BS in Mathematics	3.98 GPA	2019

SKILLS & CERTIFICATIONS

- Programming languages and tools: Python, Rust, Lean.
- Data Science Boot Camp Certification, The Erdős Institute. [2025 Certificate](#).
- Soft skills: Public speaking, technical writing, mentorship, and meeting facilitation.

SELECTED PROJECTS

Contributions to open-source software libraries – The Ethereum Foundation - Remote 2025-Present

- Contributed to [Arklib](#), an open-source library written in Lean, maintained by the Ethereum Foundation for formally verifying the security of cryptographic protocols implemented in Rust for the Ethereum blockchain. Resolved 7 open issues. [Sample PR](#).
- Contributed to [CompPoly](#), a Lean library for computable data types for building performant and verified implementations of cryptographic protocols. Implemented definitions and functions for several core data types. Resolved 12 open issues. [Sample PR](#).

Windpower Forecasting in Quebec - The Erdős Institute - Remote 2025

- Collected four years of wind power generation data from Hydro-Québec as well as weather data and designed machine learning models in Python to forecast power generation using weather, succeeding with over 80% accuracy. Results available [here](#).

Encrypted Messaging Application 2025

- Built a command line application for post-quantum secure text encryption in Rust. [Github](#).

Dissertation Research in Algebraic Geometry - The University of Texas at Austin - Austin, TX 2019-2025

- Conducted novel research in algebraic geometry that was presented at 6 international conferences and 5 invited talks at university seminars over 5 years. Research results were disseminated in 5 [articles](#); two publications and three articles in review.
- Developed techniques in tropical geometry to simplify geometric objects arising from polynomials for the purpose of computations. Generalized central theorems in the field by developing a modular framework for the proofs. Research made use of tools from topology, algebraic geometry, representation theory, and combinatorics. Resolved four unsolved open problems.

SELECTED PUBLICATIONS & PREPRINTS

- Desmond Coles. *Balancing for spherical tropical curves*. 2025. [arXiv:2505.04839](#)
- Desmond Coles and Martin Ulirsch. *Towards the tropicalization of reductive groups*. 2025. [arXiv:2503.21654](#)
- Desmond Coles and Netanel Friedenberg. *Locally finite completions of polyhedral complexes*. Disc. Comput. Geom., 2024.
- Desmond Coles and Netanel Friedenberg. *A construction of algebraizable formal models*. 2023. [arXiv:2303.13646](#)
- Desmond Coles. *Spherical tropicalization and Berkovich analytification*. Math. Z., 303(4): Paper No. 86, 23, 2023.

WORK EXPERIENCE

Assistant Professor of Instruction - The University of Texas at Austin - Austin, TX 2025-Present

- Course design, lecturing, and mentoring students. Teaching advanced calculus and number theory to 300+ students per semester.

Graduate Researcher & Teaching Assistant - The University of Texas at Austin - Austin, TX 2019-2025

- Conducted a five-year research program. Organized 3 seminars over 10 semesters as well as a summer long multi-university program for 100+ PhD students, postdocs, and professors to learn novel techniques in their field. [CV](#).
- Designed and taught classes with 100+ students for 12 semesters and mentored advanced students through 1-on-1 courses.

Teaching Assistant & Tutor - The Ohio State University - Columbus, OH 2015-2019

- Designed and taught classes of 30+ undergraduate students for 6 semesters. Tutored advanced mathematics courses.

LEADERSHIP

Graduate Representative Committee - The University of Texas at Austin - Austin, TX 2020-2022

- Founded the Graduate Representative Committee for UT Austin math department. Advocated for 100+ PhD students regarding university policy. Designed administrative processes and met regularly with university administration.

COVID-19 Outreach - The University of Texas at Austin - Austin, TX 2020-2022

- Organized an outreach program with UTeach to support 100+ students in public schools during the COVID-19 pandemic. Designed educational materials, coordinated volunteers, coordinated with school administration, and acquired funding.